

Artificial Intelligence

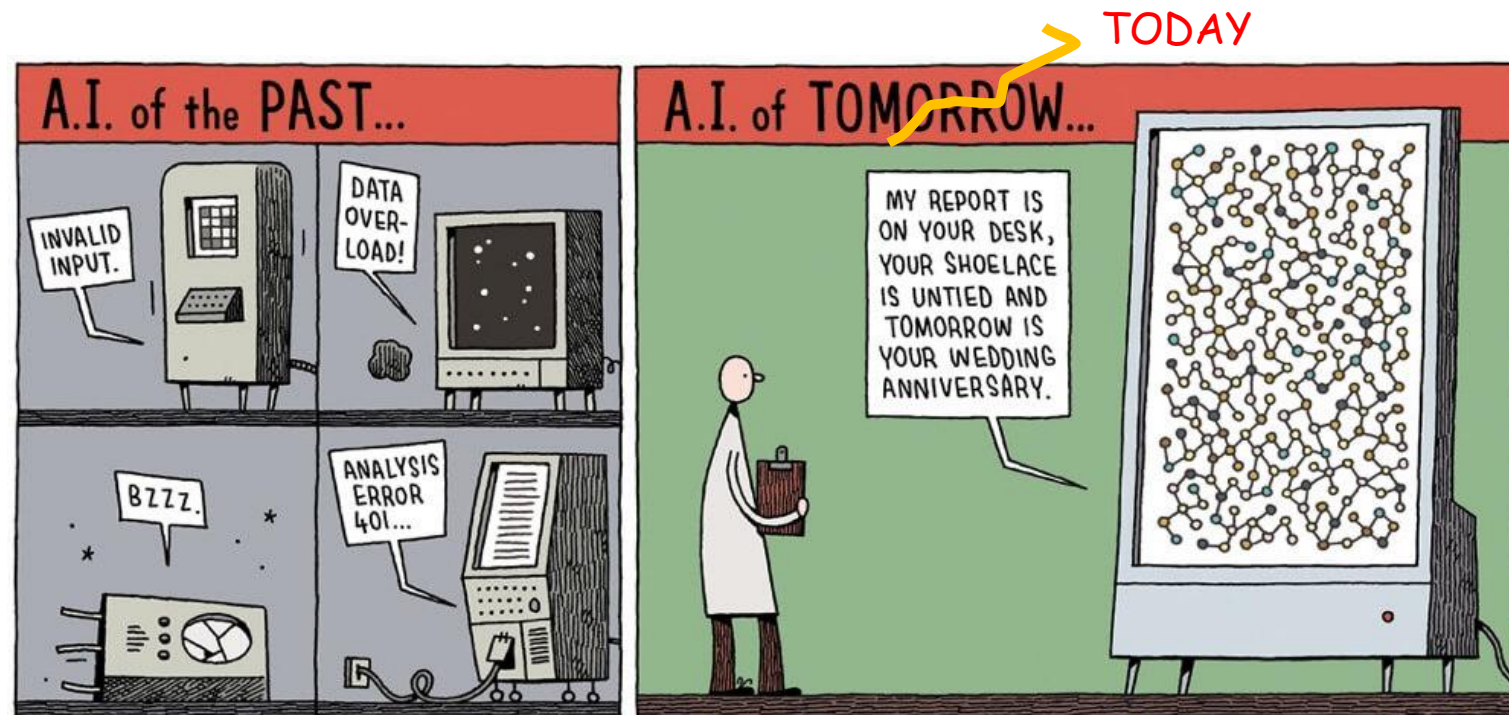
An Advanced Course About Foundations

Introduction

Marco Piastra

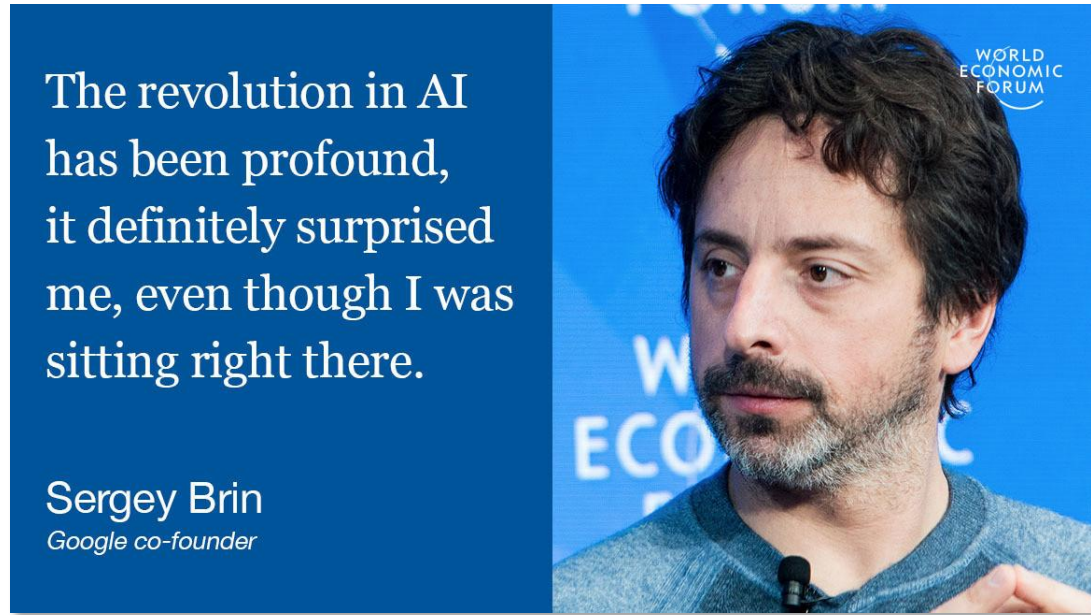
Prologue: Apropos Artificial Intelligence

Artificial Intelligence: *then and now*



[Image from <https://www.tomgauld.com/portfolio>]

AI revolution?



- **Sergey Brin** [Google Co-Founder, January 2017]

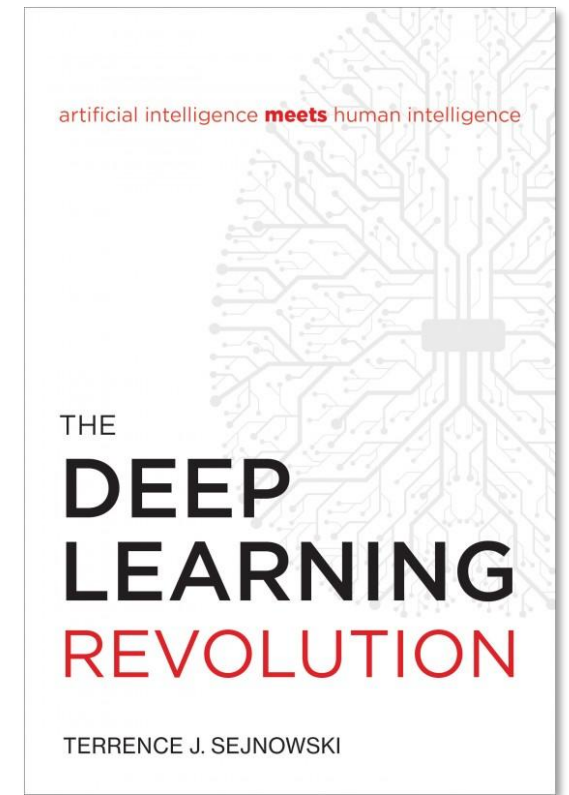
*"I didn't pay attention to it [i.e. **Artificial Intelligence**] at all, to be perfectly honest."*

"Having been trained as a computer scientist in the 90s, everybody knew that AI didn't work."

*People tried it, they tried **neural nets** and none of it worked."*

[Quote and image from <https://www.weforum.org/agenda/2017/01/google-sergey-brin-i-didn-t-see-ai-coming/>]

The Deep Learning Revolution (in its early days)



- **Terrence J. Sejnowski** [President of the Neural Information Processing (NIPS) Foundation, October 2018]

"How deep learning—from Google Translate to driverless cars to personal cognitive assistants—is changing our lives and transforming every sector of the economy."

"AI is now awakening and transforming our world.

Driving these breakthroughs is the deep learning revolution [...]"

[Quote and image from <https://mitpress.mit.edu/books/deep-learning-revolution>]

Artificial Intelligence hysteria?



AI's current hype and hysteria could set the technology back by decades

July 24, 2019 10:11am BST

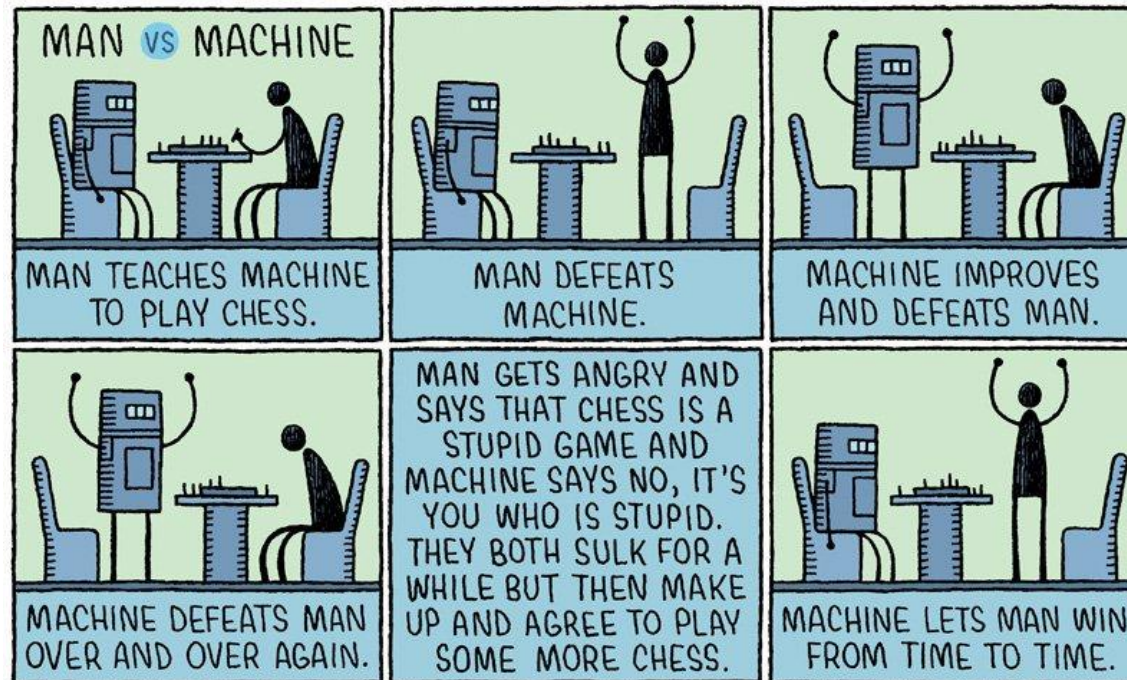
AI isn't as scary as we imagine. AndreyZH/Shutterstock

The reality of AI is currently very different, particularly when you look at the threat of automation. Back in 2013, researchers estimated that, in the following ten to 20 years, 47% of jobs in the US could be automated. Six years later, instead of a trend towards mass joblessness, we're in fact seeing US unemployment at a historic low.

Current AI is good at **finding patterns in large datasets**, and not much else.

[Quote from <https://theconversation.com/ais-current-hype-and-hysteria-could-set-the-technology-back-by-decades-120514>]

Computers vs. Humans

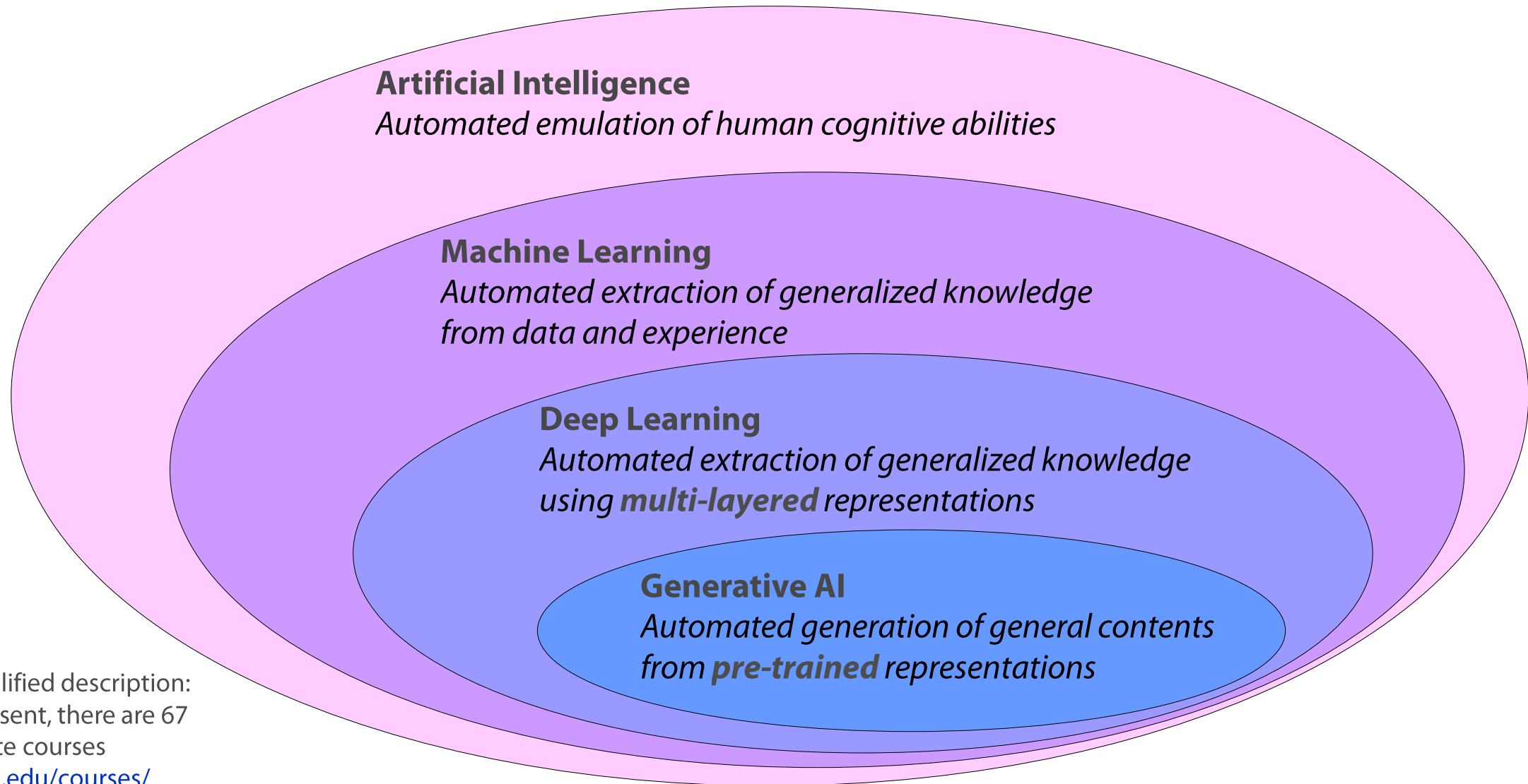


TOM GAULD

[Image from <https://www.tomgauld.com/portfolio>]

Artificial Intelligence (in a nutshell)

The Artificial Intelligence Cosmos



- ❖ This is a very simplified description: at Stanford, at present, there are 67 AI-related graduate courses <https://ai.stanford.edu/courses/>

Early Periods: Symbolic Reasoning

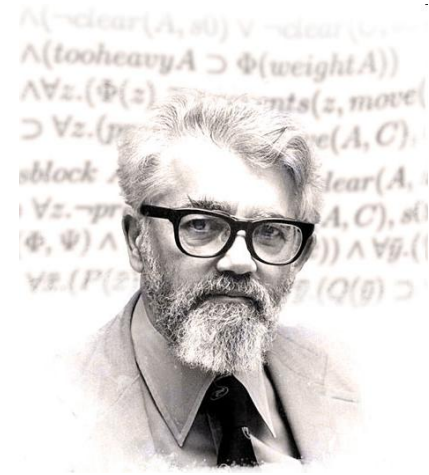
"Artificial Intelligence" (first appearance of the term)

“We propose that a two-month, ten-man study of **artificial intelligence** carried out during the summer of 1956 [...]

The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of **intelligence** can in principle be ***so precisely described*** that a machine can be made to ***simulate*** it. [...]

It may be speculated that a large part of human thought consists of manipulating **words** according to **rules** of **reasoning** and **rules** of **conjecture**."

[John McCarthy et al., 1955 – *all emphases added*]



[Image from Wikipedia]

Automated Symbolic Logic

*“The only way we know of expressing abstractions [...] is in language.
That is why we have decided to program a system which reasons verbally. ”*

[John McCarthy, PROGRAMS WITH COMMON SENSE, 1959]

■ **Formal, Symbolic Logic as a viable candidate**

In those days, Formal Logic seemed an obvious choice

- its formalism is derived from *natural language*
(since G. Frege, 1879)
- it has a clear semantics
(since ‘The Semantic Theory of Truth’, A. Tarski, 1930)
- it is *compositional*
(the meaning of a complex expression is determined by its constituent and the rules used to combine them)
- it can be turned into a *computing system*
(see Herbrand’s Theorem, 1930)

Automated Symbolic Logic

*“The only way we know of expressing abstractions [...] is in language.
That is why we have decided to program a system which reasons verbally. ”*

[John McCarthy, PROGRAMS WITH COMMON SENSE, 1959]

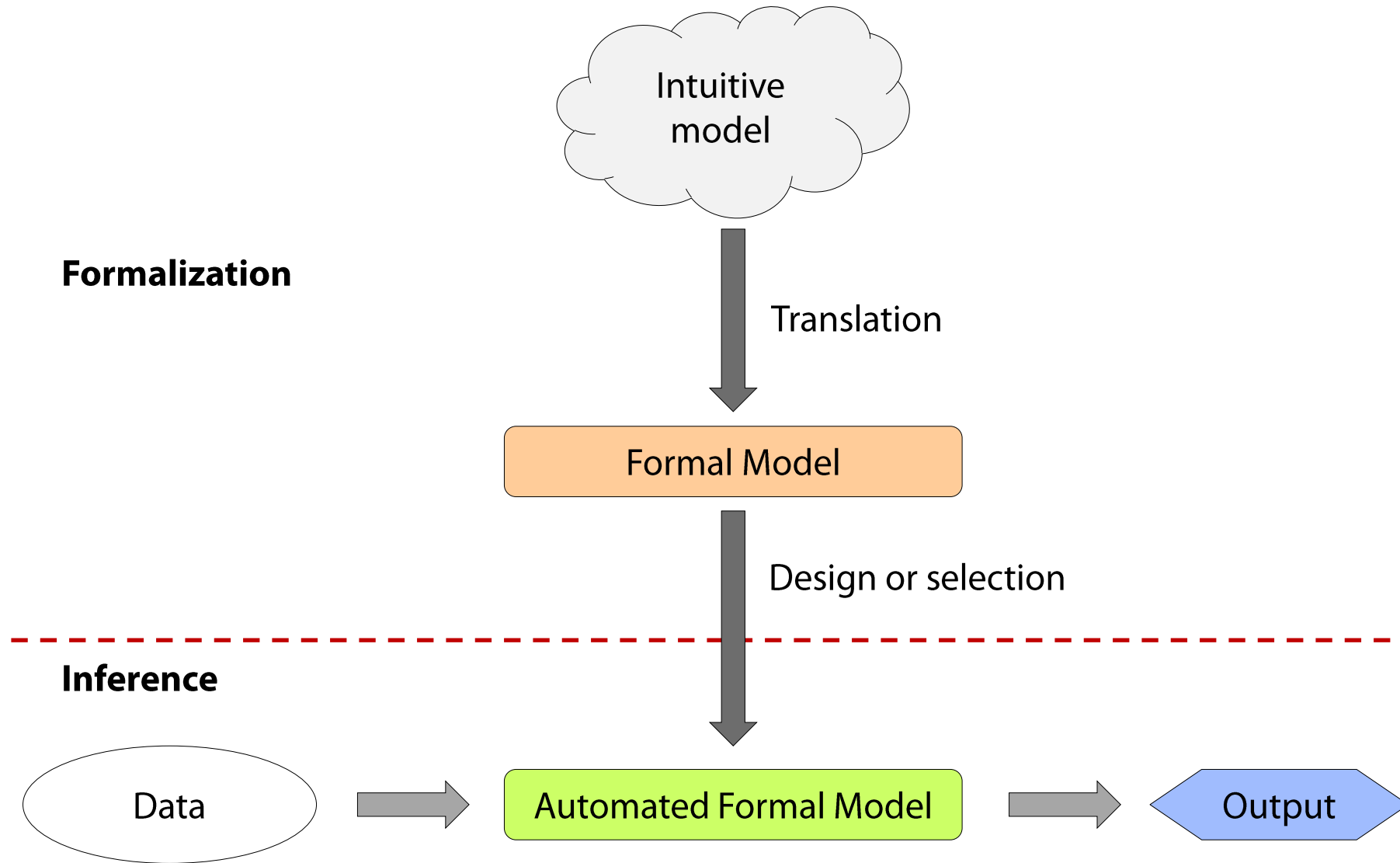
■ **Logicism dominated the early period of AI**

The logical approach in three theses:

(from ‘Logic and artificial intelligence’, N. J. Nilsson, 1991 – *emphasis added*)

1. Intelligent machines will have **knowledge** of their **environments**.
2. The most versatile intelligent machines will represent much of their knowledge about their environments **declaratively**.
3. For the most versatile machines, the language in which declarative knowledge is represented must be at least as expressive as **first-order predicate calculus** [i.e. **symbolic logic**].

AI Symbolic Framework



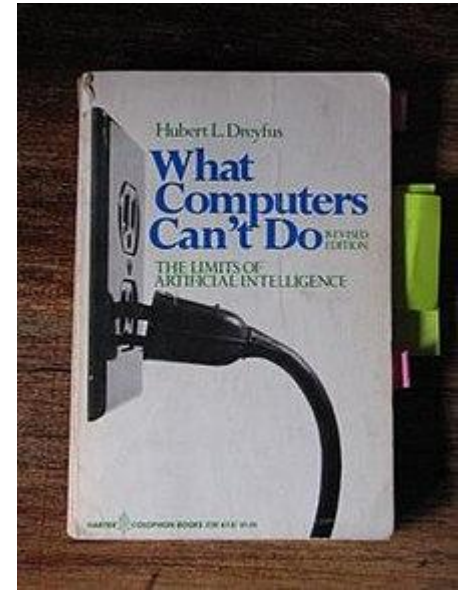
An Alternative Approach: Connectionism

Criticism of the symbolic approach

[H. Dreyfus, 1972]

Human intelligence and expertise depend primarily on ***unconscious*** processes rather than *conscious* symbolic manipulation, and that these *unconscious* skills can never be fully captured in formal rules.

(These ideas are now embedded in the so-called ***sub-symbolic*** approach)



[Image from Wikipedia]

'Out of Control', BBC2 Horizon documentary, 2012

- *How big is the unconscious mind (in humans)?*

In this BBC documentary, several senior neuro-scientists were asked to represent on a sheet of paper the extent of **conscious activity** (vs. **unconscious**) in the human brain



[Image from <https://www.bbc.co.uk/programmes/p00pyhx2>]

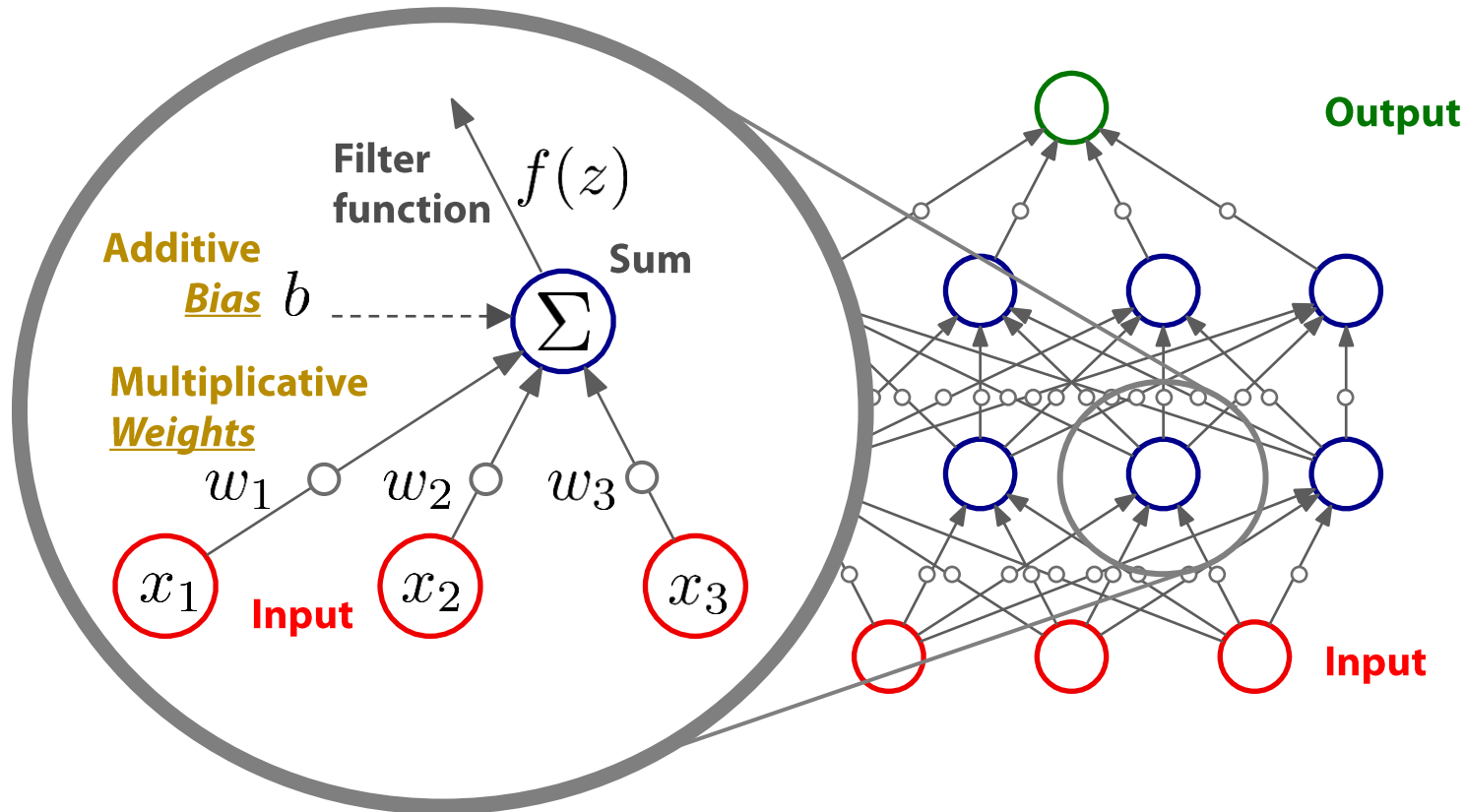
Artificial Neural Networks: At the Core of AI Revolution

An assembly of simple computational units



[Images from Wikipedia]

[Rumelhart, D.E., J.L. McClelland 1986]

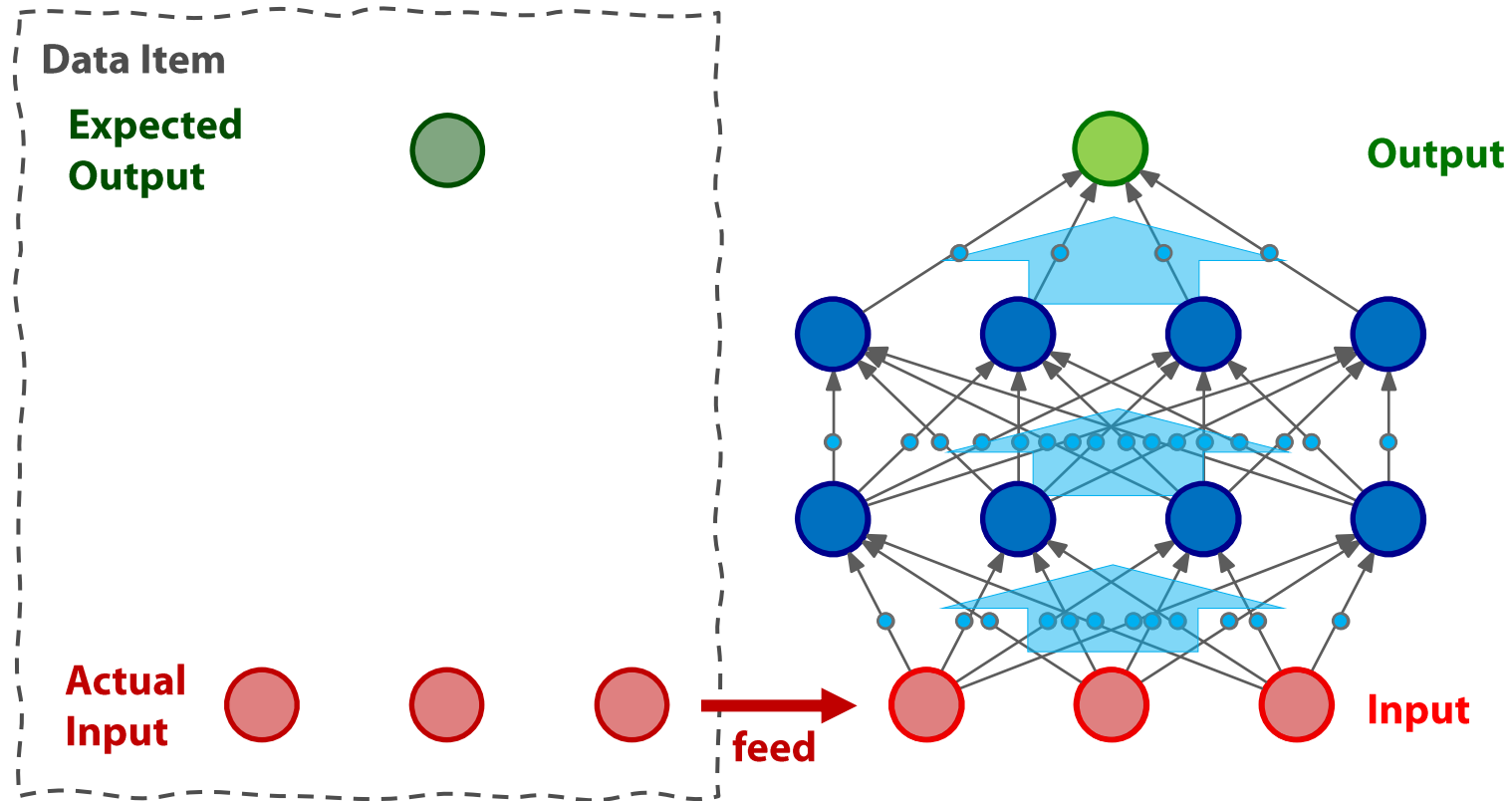


*Each unit performs
simple arithmetic
operations*

*Weights and Biases
are the only
mutable parts*

Learning from Data

Incremental, numerical optimization

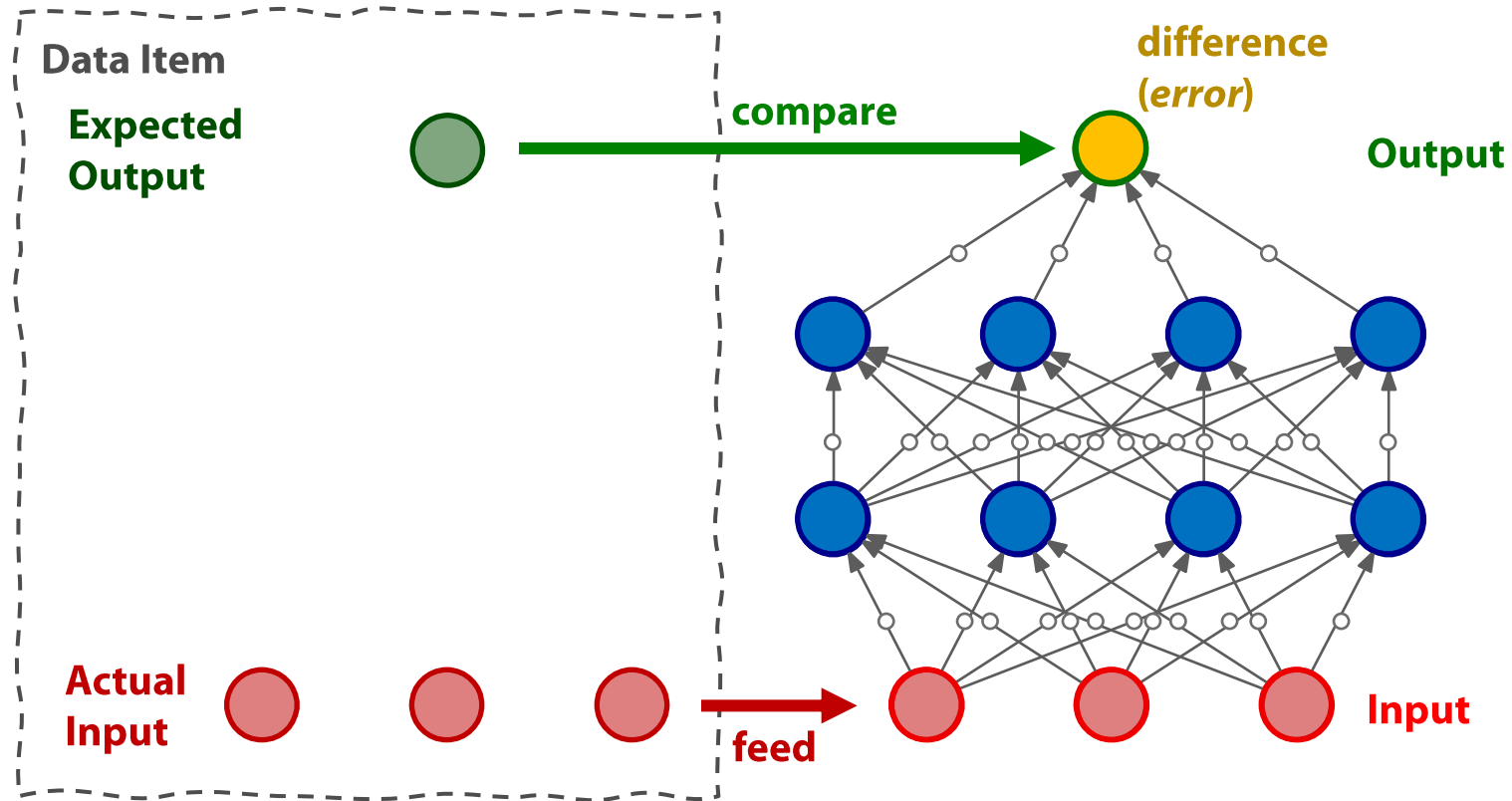


*Learning from
data items:
input + output*

*Input is fed to the
network and the
output is computed*

Learning from Data

Incremental, numerical optimization

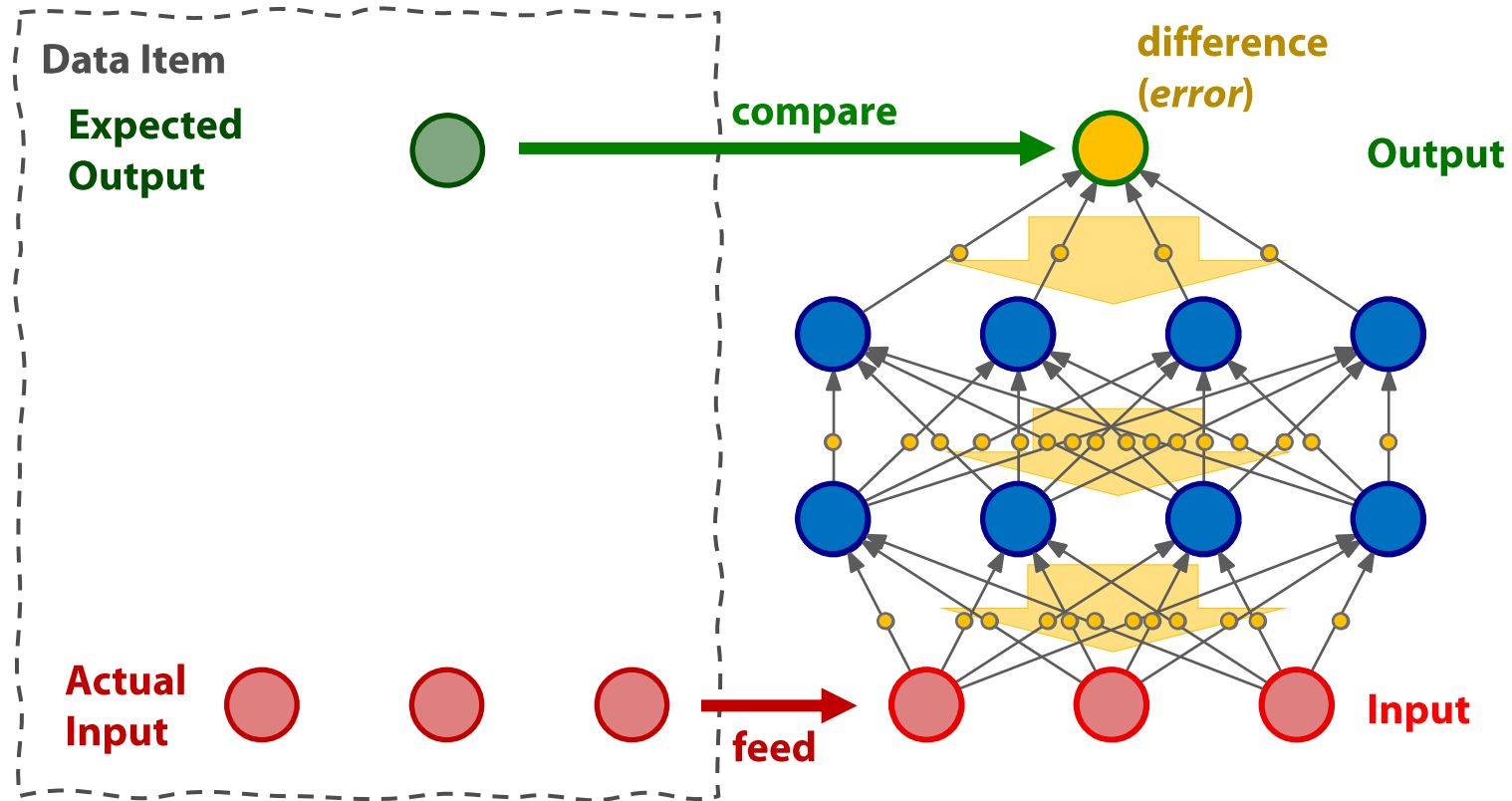


Output is compared with expected value

Error is estimated from difference

Learning from Data

Incremental, numerical optimization

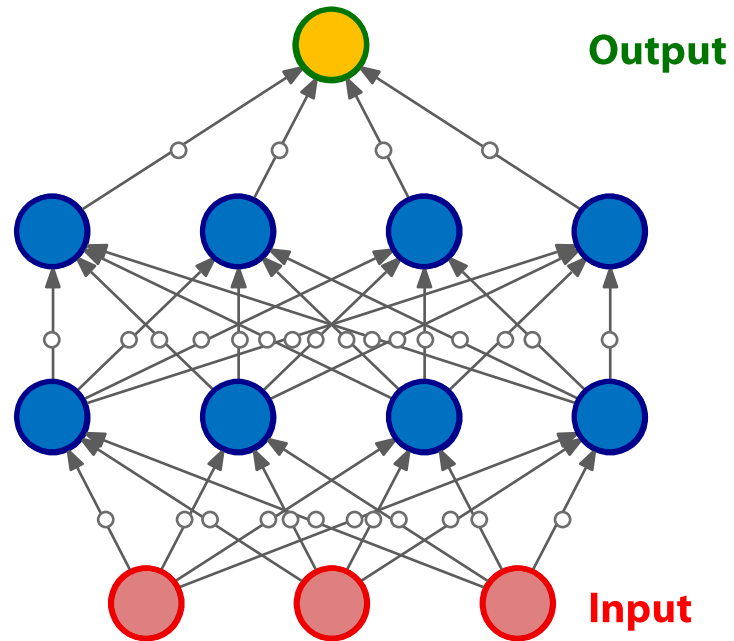
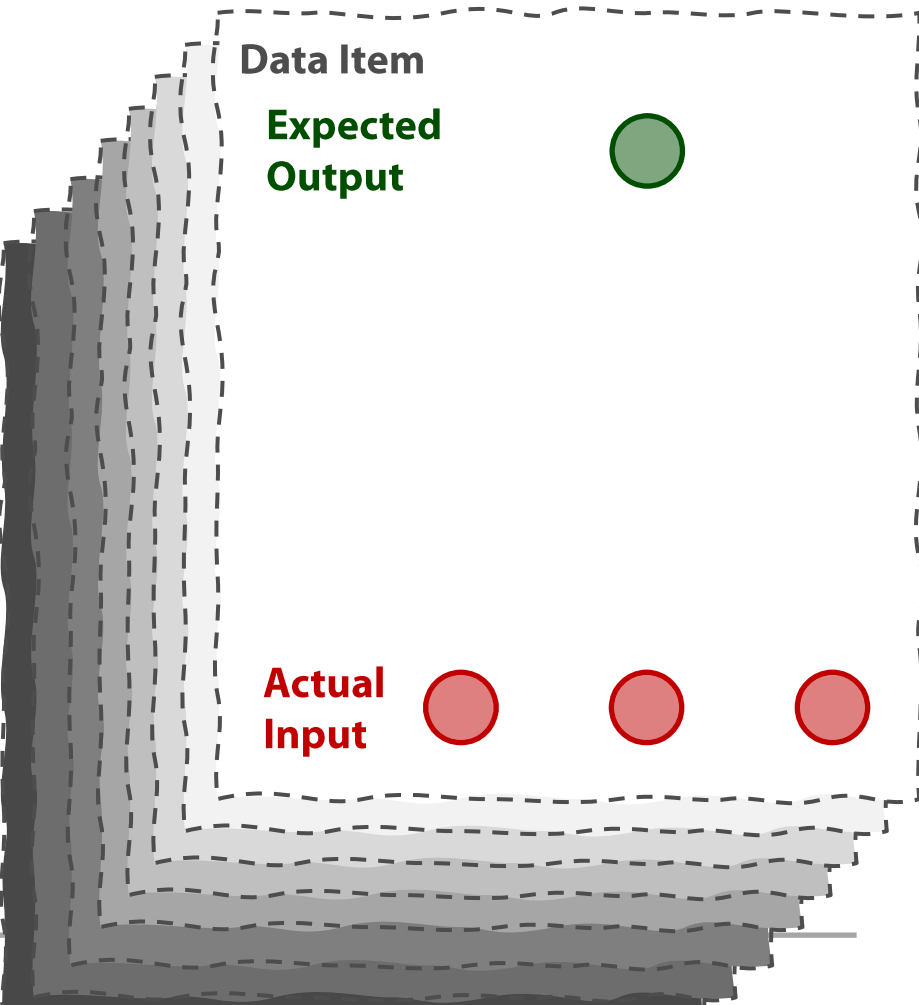


*Error is propagated
in the opposite
direction*

*To adapt gradually all
parameters*

Learning from Data

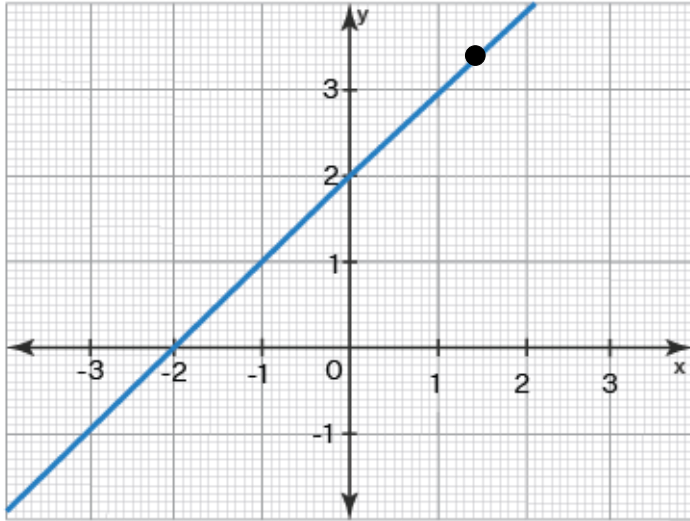
Incremental, numerical optimization



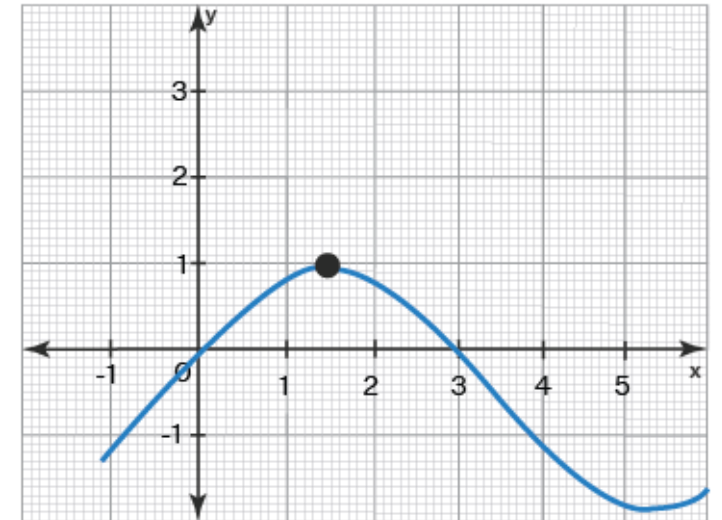
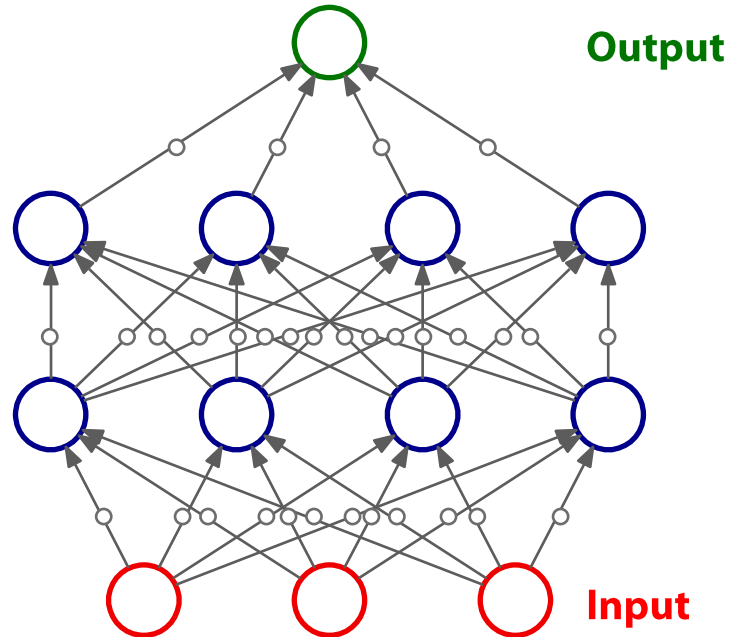
The process is repeated on huge datasets

Non-Linear Functions

Under mild mathematical constraints,
an artificial neural network can represent any non-linear function

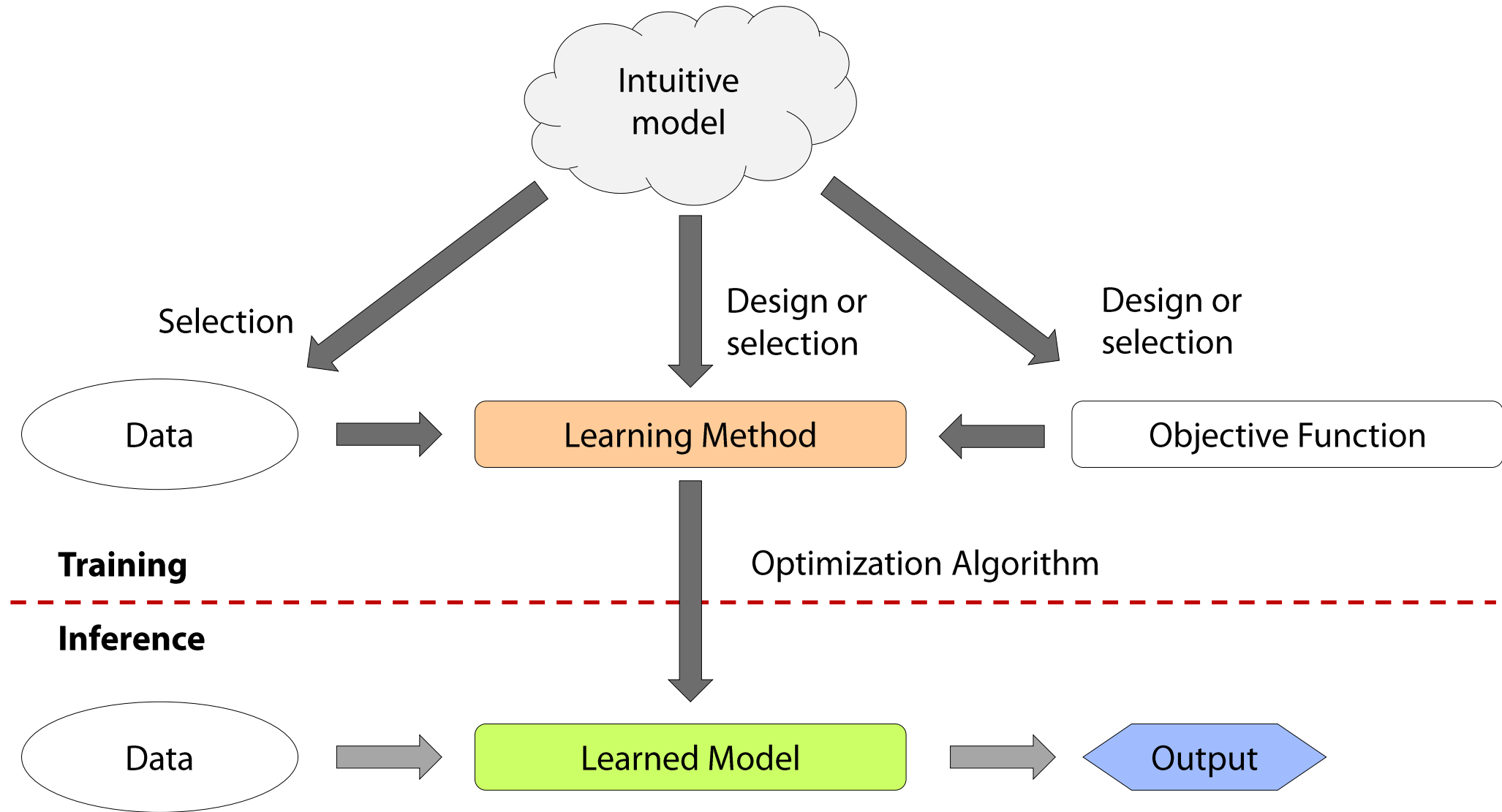


Linear function



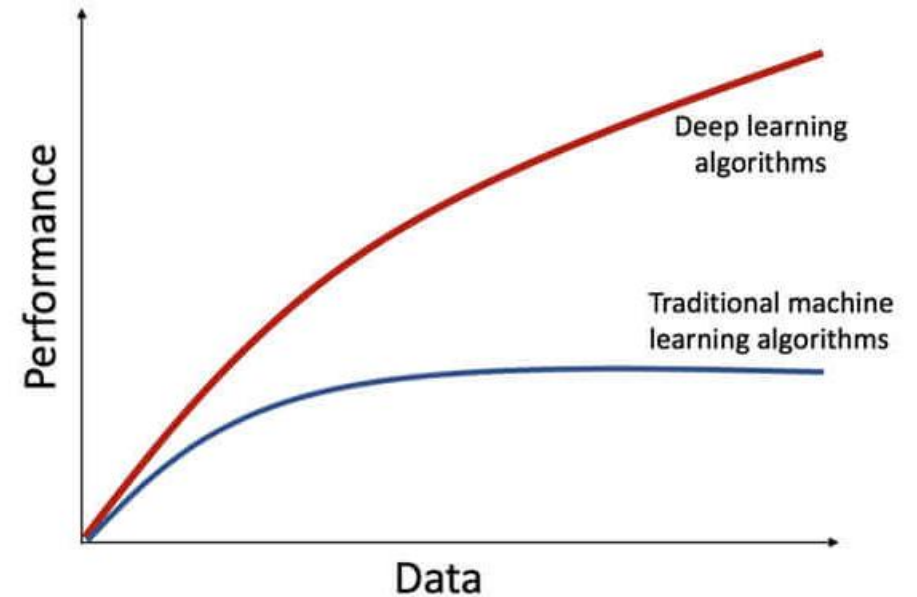
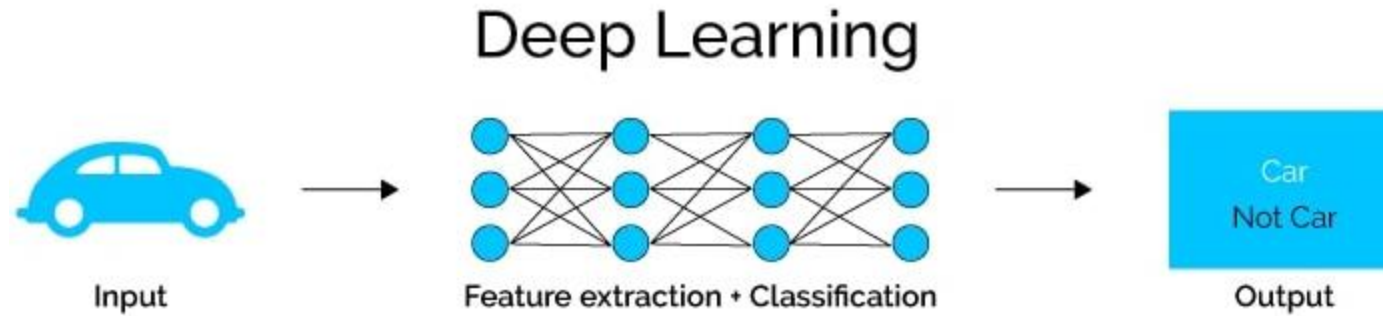
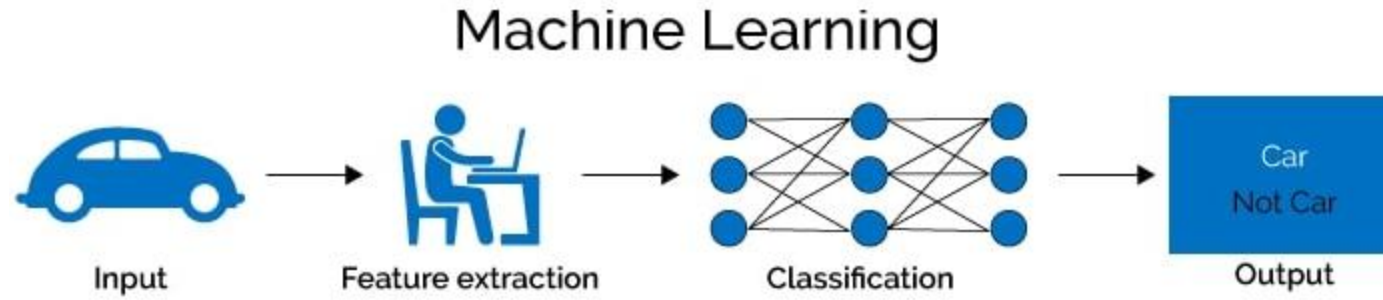
Non-linear function

Machine/Deep Learning



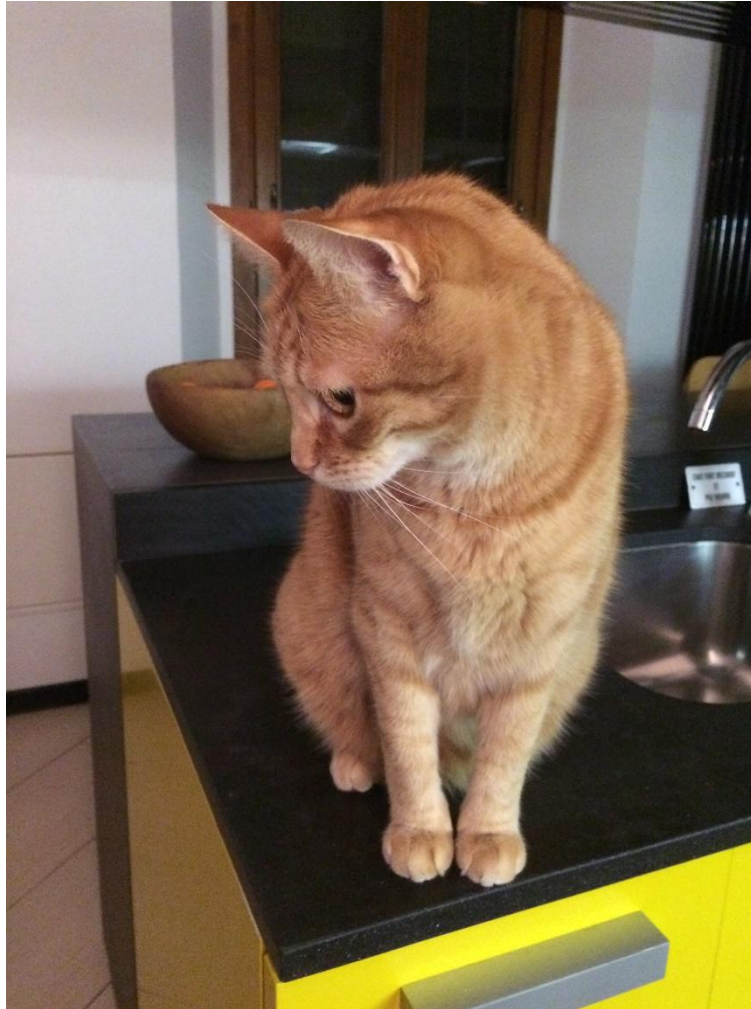
AI Revolution: Deep Neural Networks

Machine Learning vs Deep Learning

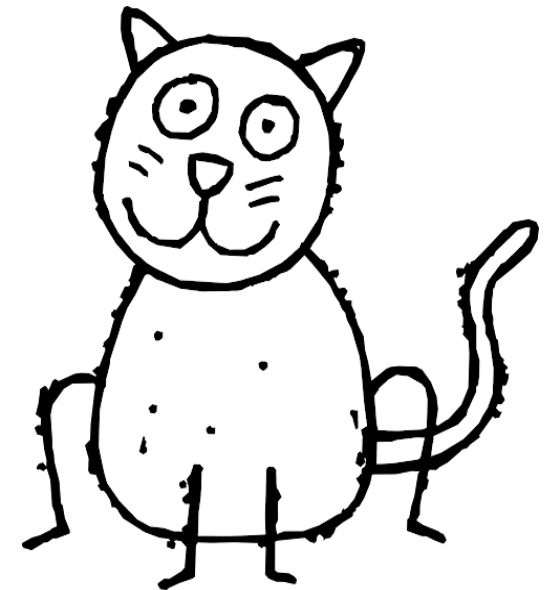


Deep Learning and Artificial Perception

*Is there a cat
in this picture?*

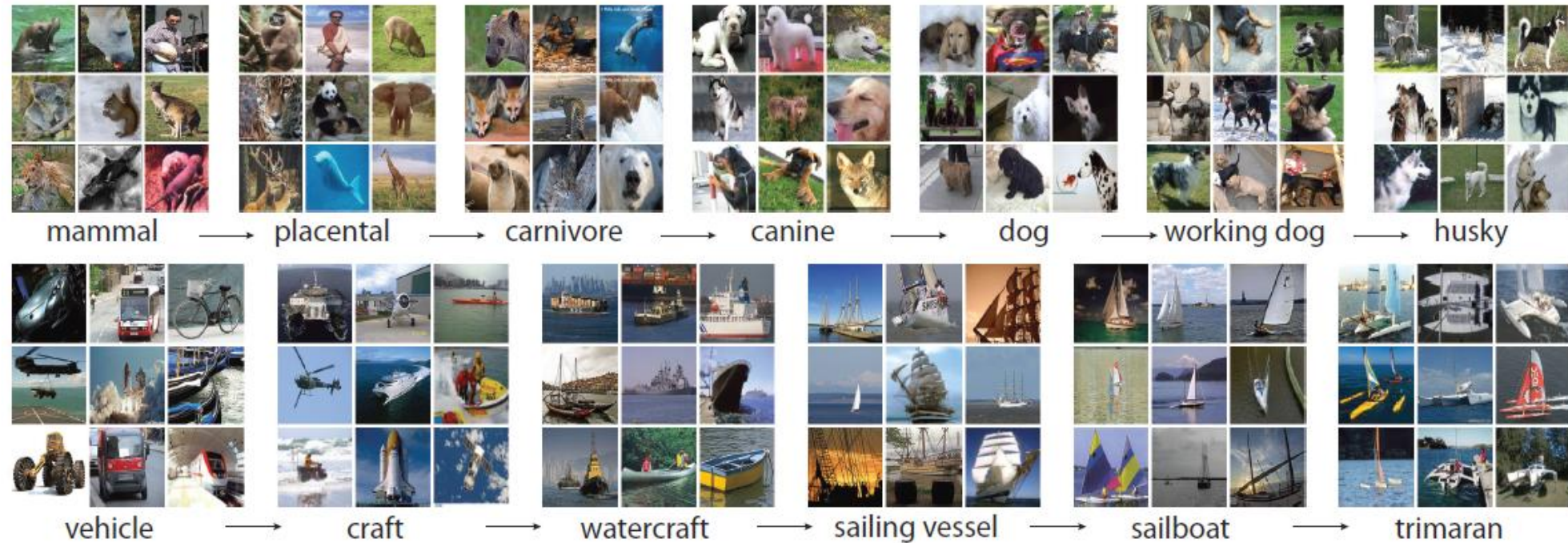


[this is *my* cat, Rabarbaro]



ImageNet Challenge

- The ImageNet Large Scale Visual Recognition Challenge



1,461,406 full resolution images

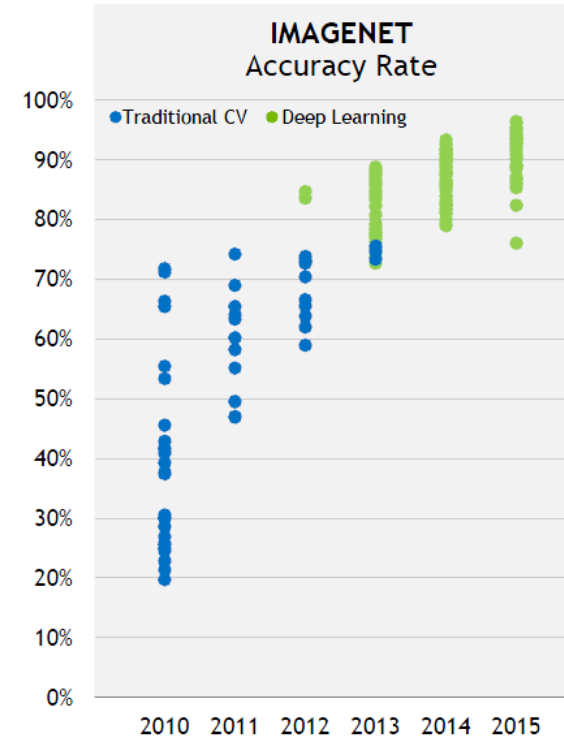
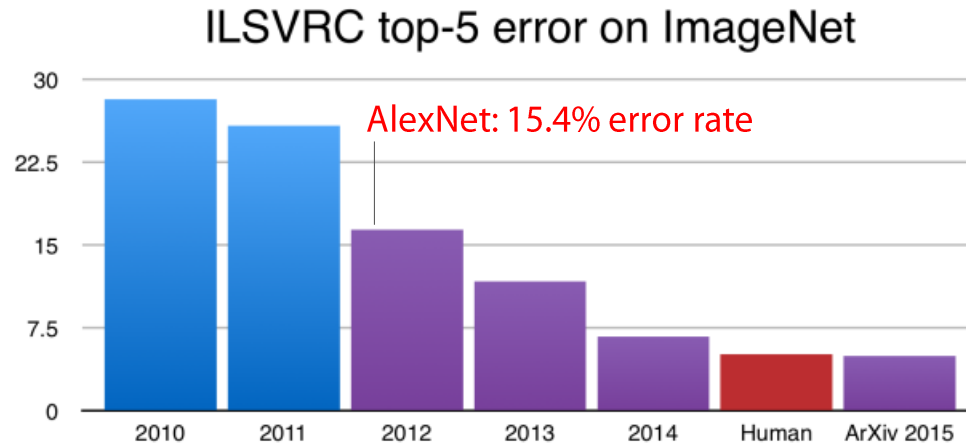
Complex and multiple textual annotation,
hierarchy of 1000 object classes along several dimensions

The image classification challenge is run annually since 2010

[figures from www.nvidia.com]

ImageNet Challenge

- The ImageNet Large Scale Visual Recognition Challenge



1,461,406 full resolution images

Complex and multiple textual annotation,
hierarchy of 1000 object classes along several dimensions

The image classification challenge was run annually from 2010 to 2017

[figures from www.nvidia.com]

Generative AI: ChatGPT

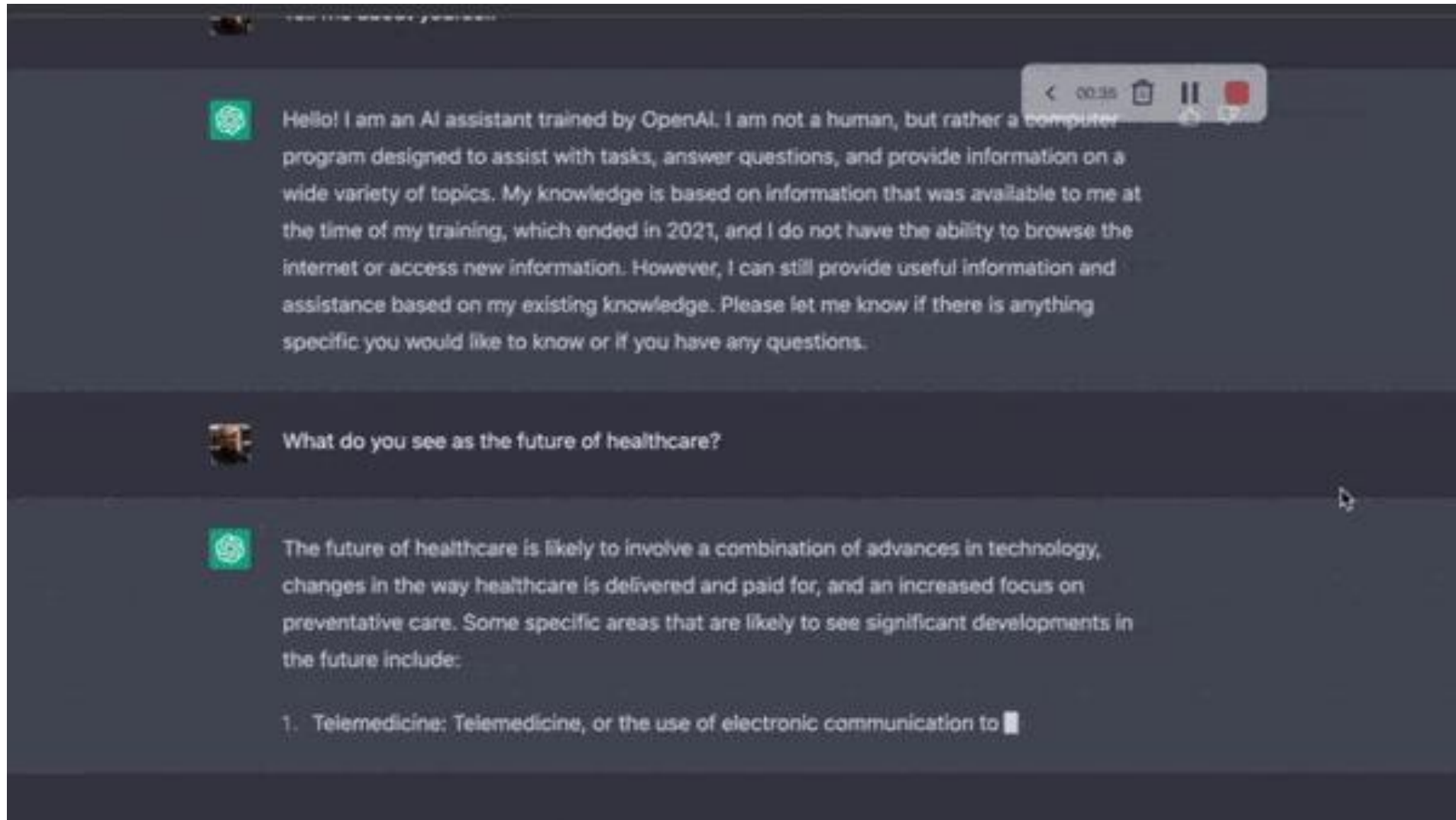


Image from <https://info.homecareinnovationforum.com/the-potential-of-chatgpt-in-home-health-care>

Language is not the only generative way

- **DALL-E**

Image generation from text prompts

«A teapot in the shape of an avocado»

[Image from <https://www.nytimes.com/2022/04/06/technology/openai-images-dall-e.html>]



Generative AI

ChatGPT is not alone

■ SORA

Generating videos from text prompts

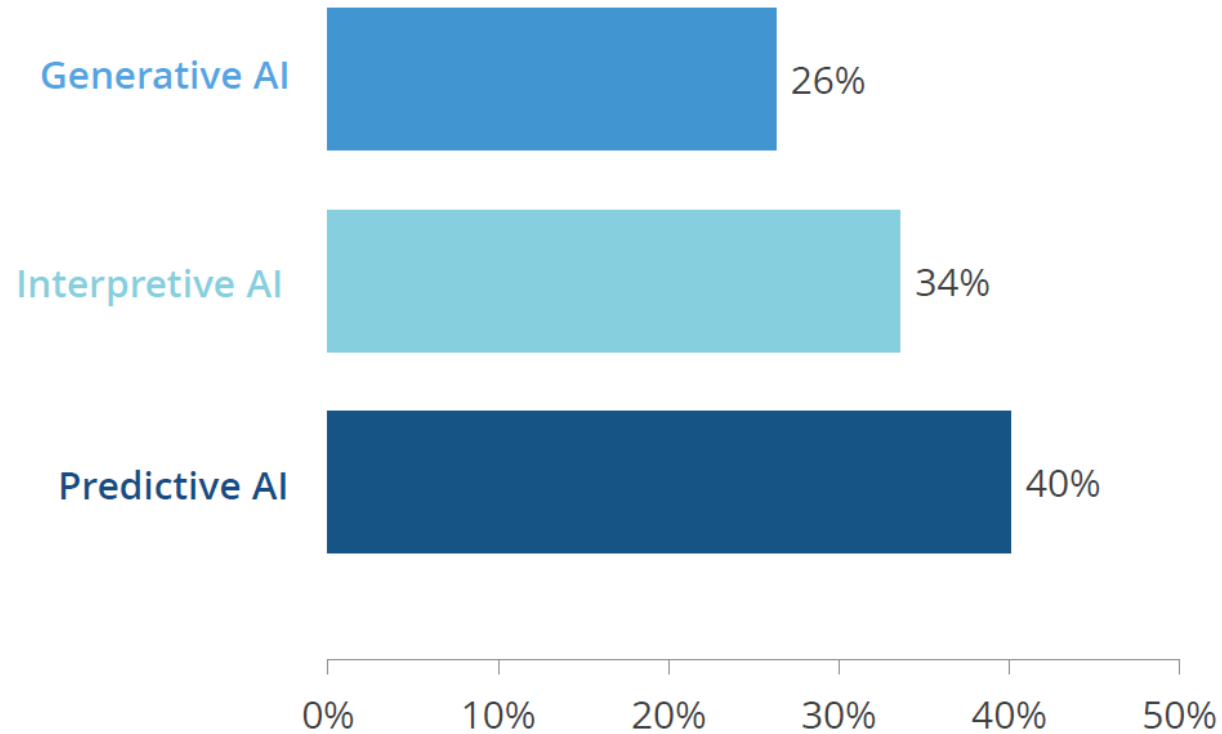


[Video clip from <https://openai.com/index/sora/>]

«A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.»

Generative AI is Just ONE Technology

“What is the best estimate of how your total AI-related investments will be divided among the following categories over the next 18 months?”



Source: EMEA AI, Data, Analytics Survey, 2024 Total sample: N=920

The Nobel Prize in Chemistry 2024

They cracked the code for proteins' amazing structures

The Nobel Prize in Chemistry 2024 is about proteins, life's ingenious chemical tools. David Baker has succeeded with the almost impossible feat of building entirely new kinds of proteins. Demis Hassabis and John Jumper have developed an AI model to solve a 50-year-old problem: predicting proteins' complex structures. These discoveries hold enormous potential.

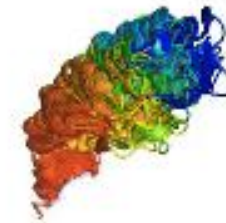
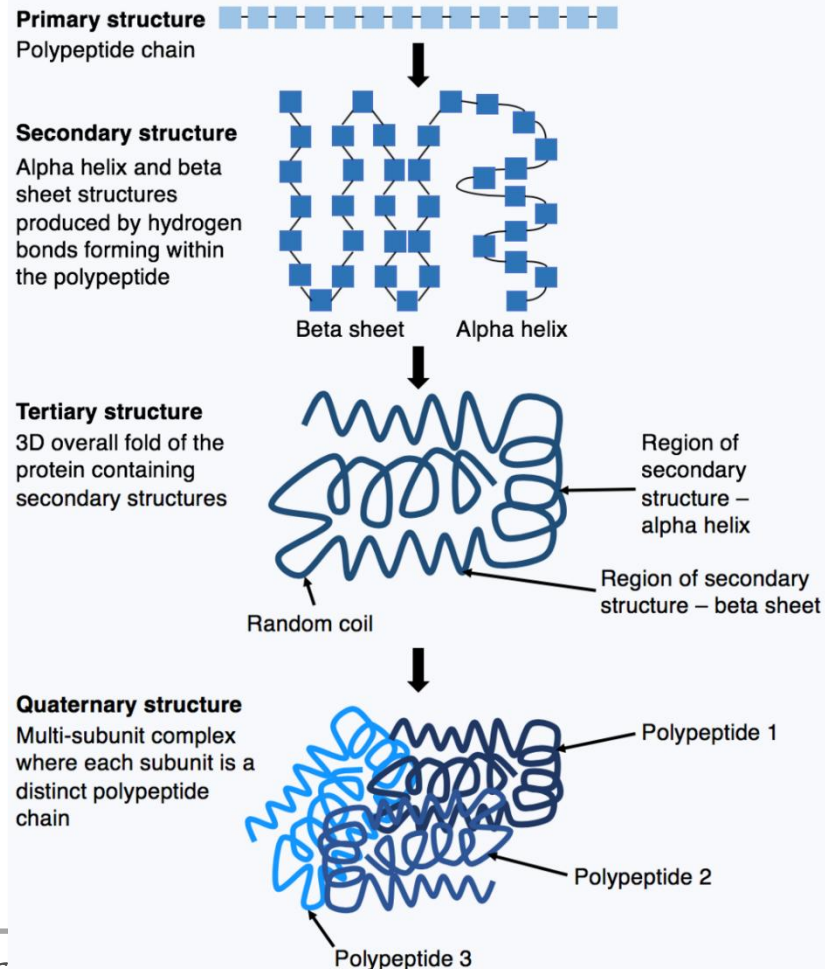


© Johan Jarnestad/The Royal Swedish Academy of Sciences

Predictive AI

Predicting 3D protein folding from primary structure

■ AlphaFold



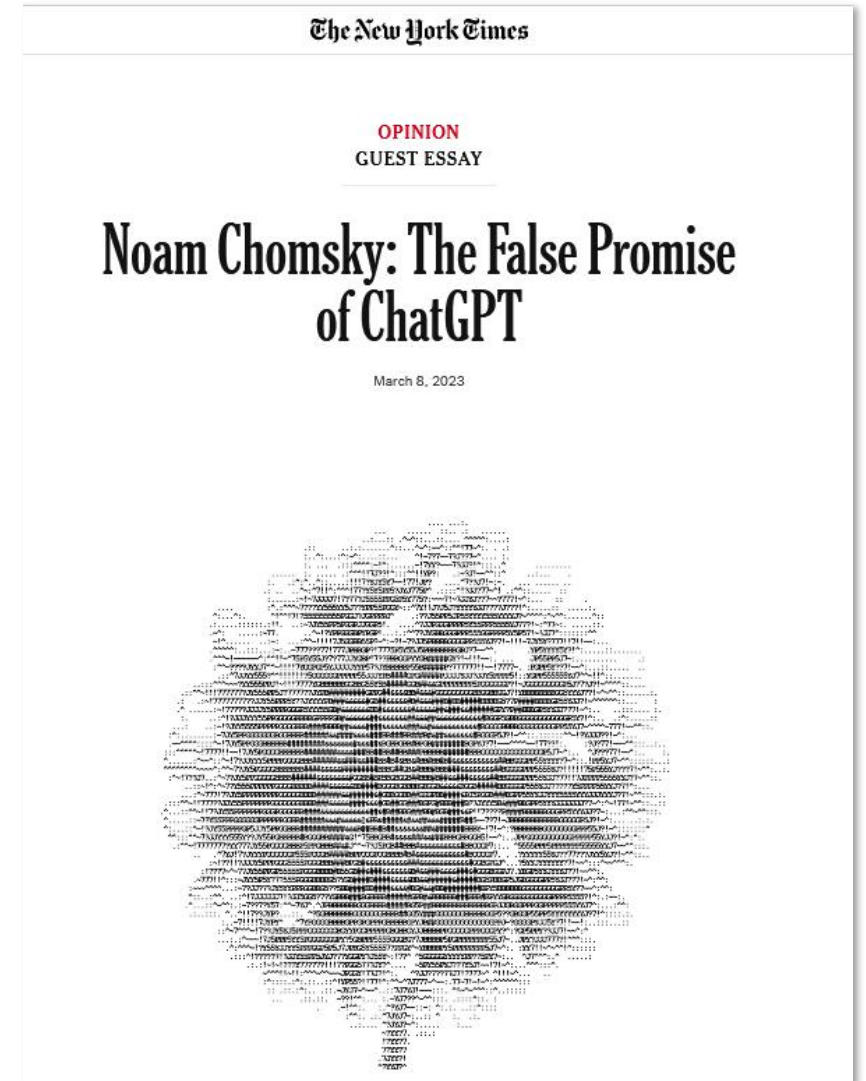
Recycling iteration 0, block 03
Secondary structure assigned from the final prediction

[Video clip from: <https://www.nature.com/articles/s41586-021-03819-2>]

Is Artificial Intelligence: Risky? Helpful?

Is Artificial Intelligence Intelligent?

“OpenAI’s ChatGPT, Google’s Bard and Microsoft’s Sydney are marvels of machine learning. Roughly speaking, they take huge amounts of data, **search for patterns in it** and become increasingly proficient at generating statistically probable outputs — such as seemingly humanlike language and thought.”



[Quote from <https://www.nytimes.com/2023/03/08/opinion/noam-chomsky-chatgpt-ai.html>]

Is Artificial Intelligence Biased?

The New York Times

Artificial Intelligence › An Unsettling Chat With Bing Read the Conversation How Chatbots Work Spotting A.I.-Generated Text

Google Researcher Says She Was Fired Over Paper Highlighting Bias in A.I.

Timnit Gebru, one of the few Black women in her field, had voiced exasperation over the company's response to efforts to increase minority hiring.

[Give this article](#) [Share](#) [Bookmark](#) [Comments 276](#)



Timnit Gebru, a respected researcher at Google, questioned biases built into artificial intelligence systems. Cody O'Loughlin for The New York Times

 By Cade Metz and Daisuke Wakabayashi

Dec. 3, 2020

[Quote from <https://www.nytimes.com/2020/12/03/technology/google-researcher-timnit-gebru.html>]

Is Artificial Intelligence Helpful?

**The
Guardian**

It's cheap, quick and available 24/7, but is a chatbot therapist really the right tool to tackle complex emotional needs?

by Alice Robb

Illustration: Javier Jaén/The Guardian

Sat 2 Mar 2024 17.00 CET



Analyse this: a therapy special

'He checks in on me more than my friends and family': can AI therapists do better than the real

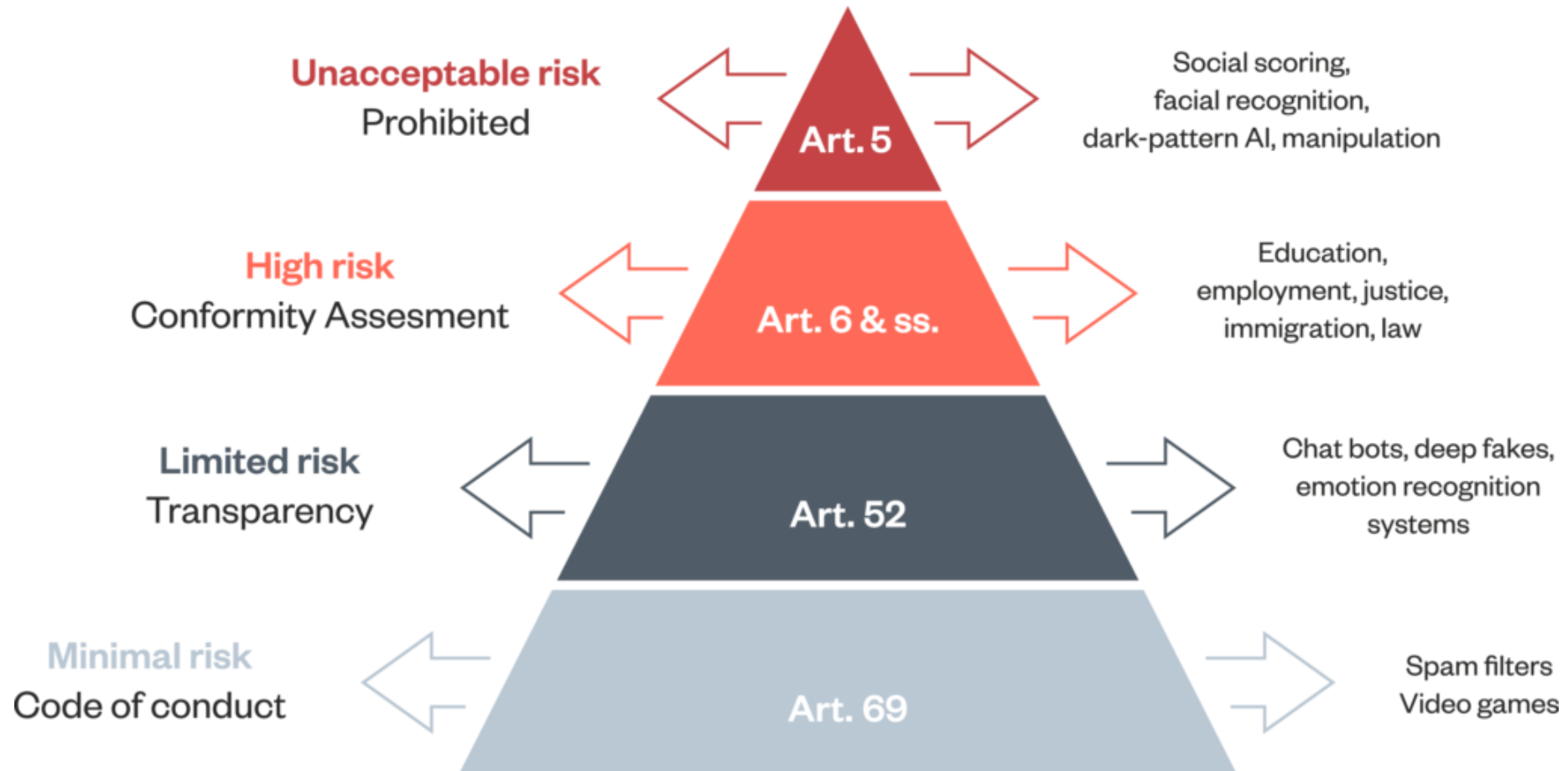
[Quote from <https://www.theguardian.com/lifeandstyle/2024/mar/02/can-ai-chatbot-therapists-do-better-than-the-real-thing>]

The EU AI Act



A risk-based approach

Regulation - EU - 2024/1689, effective since 01/08/2024



[Image from: <https://www.adalovelaceinstitute.org/resource/eu-ai-act-explainer/>]



Conformity Assessment and Self-Assessment

Quality management system

Adequate technical documentation

Consistency of AI systems and their post-market monitoring with documentation

Human Oversight

To provide authoritative safeguards against the negative effects of AI systems

Transparency Obligations

AI systems deployers should reasonably understand the system's functioning and output

Technical documentation covering AI system training, testing, and evaluation processes

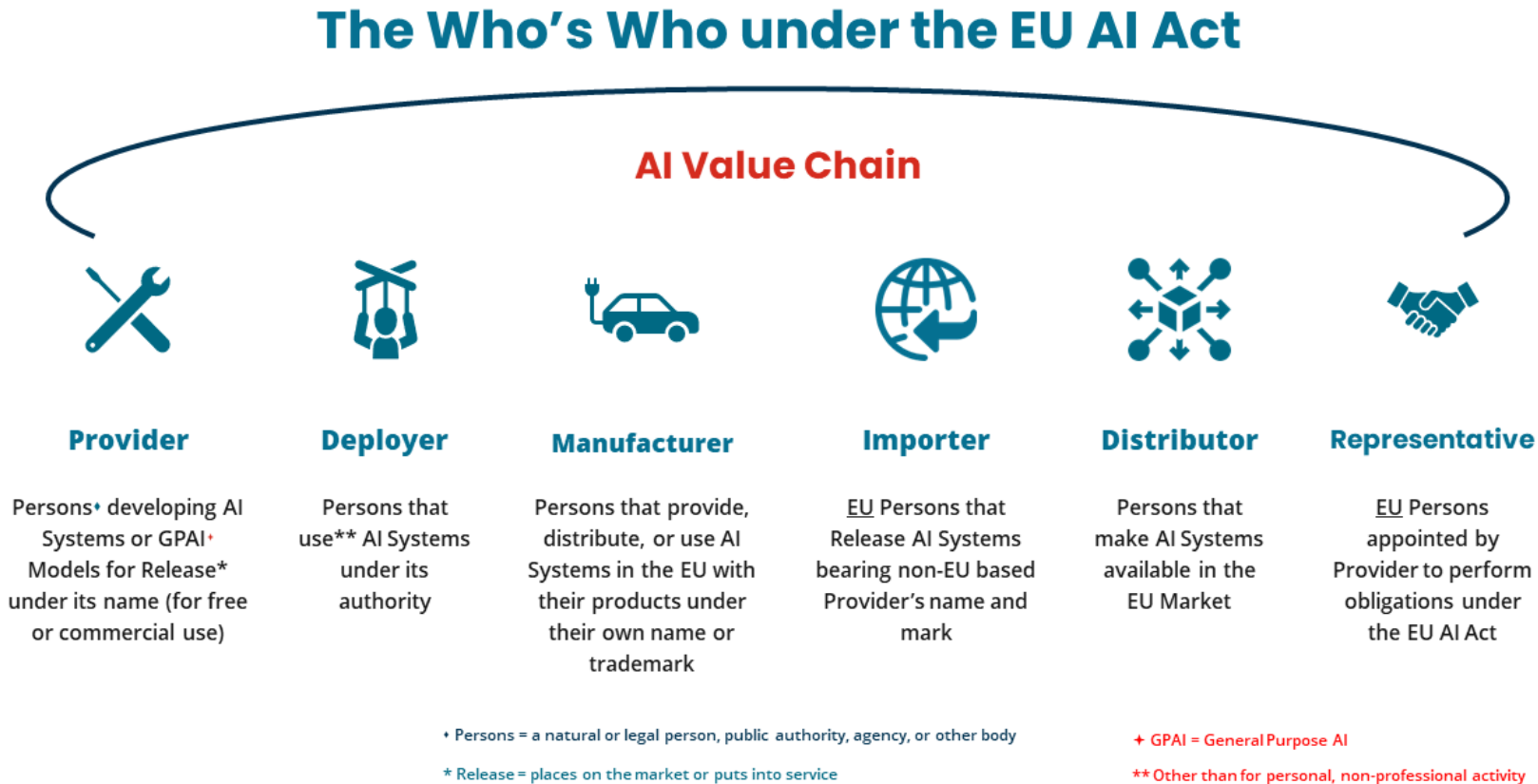
Interplay with GDPR

GDPR prohibits from processing “personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation”



Distributed responsibility

Along the entire value chain



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[Image from: <https://www.bakerdonelson.com/whos-who-under-the-eu-ai-act-spotlight-on-key-actors>]

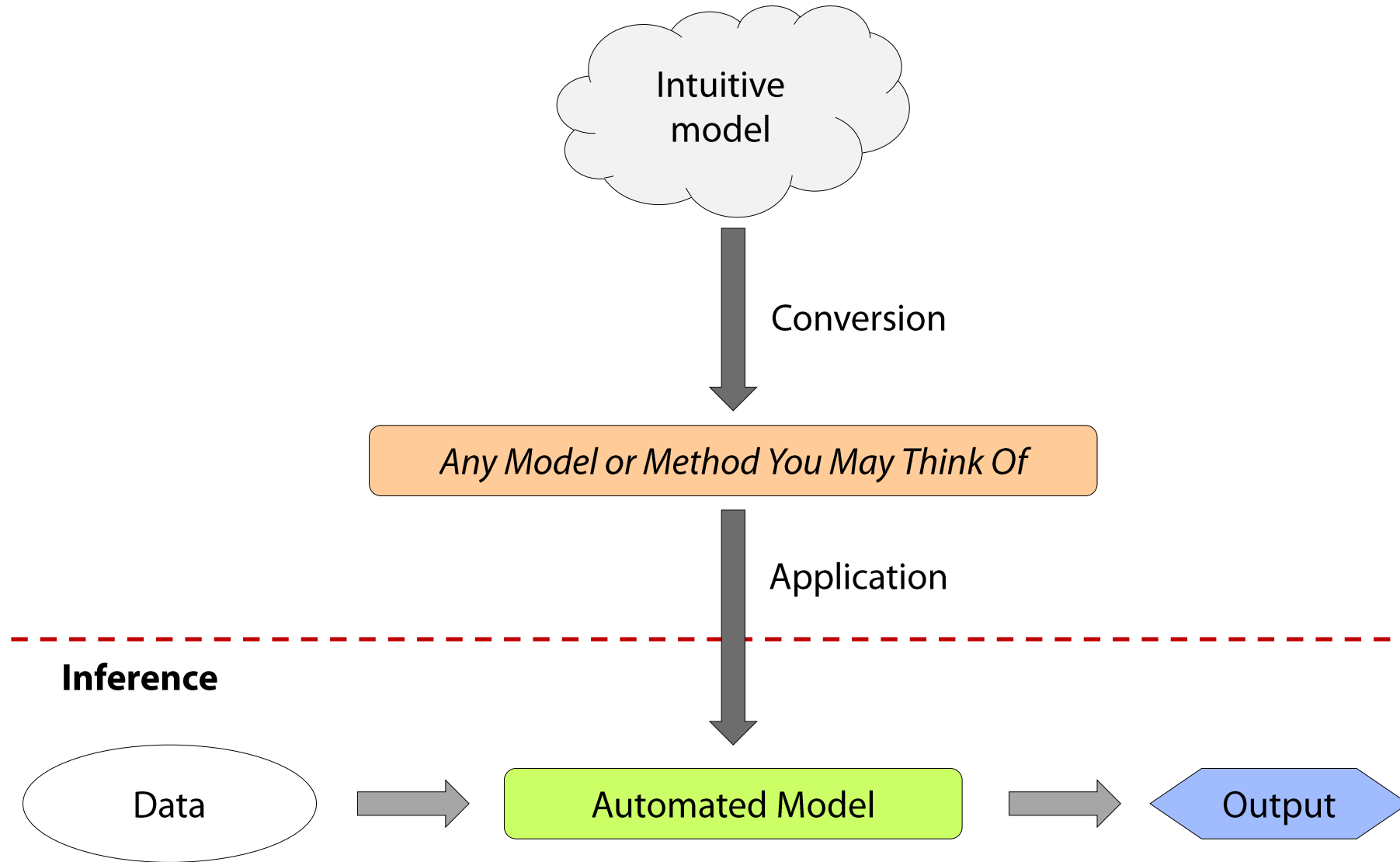
– pause –

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OK, what is this course about?

*"Inference "
(the very idea)*

Inference is a Fundamental AI Trait



What Is Inference?

- **From a theoretical standpoint**

What is the theoretical nature of *relations* between inputs and outputs?
(*Can it be 'so precisely described' – see John McCarthy?*)

What are the justifications, strength and limitations of such relations?

Can these relations be computed, at all?

- **From the point of view of computer engineering**

To what extent, if any, can inference be translated into a computer *algorithm*?

Are there practical limitations (computational complexity)?

If so, what kind of compromises need we adopt to make them *computationally feasible*?

*An aside question:
What is an algorithm, precisely?*

Turing Machine (A. Turing, 1937)

- Informal description (*more to come, later on*)

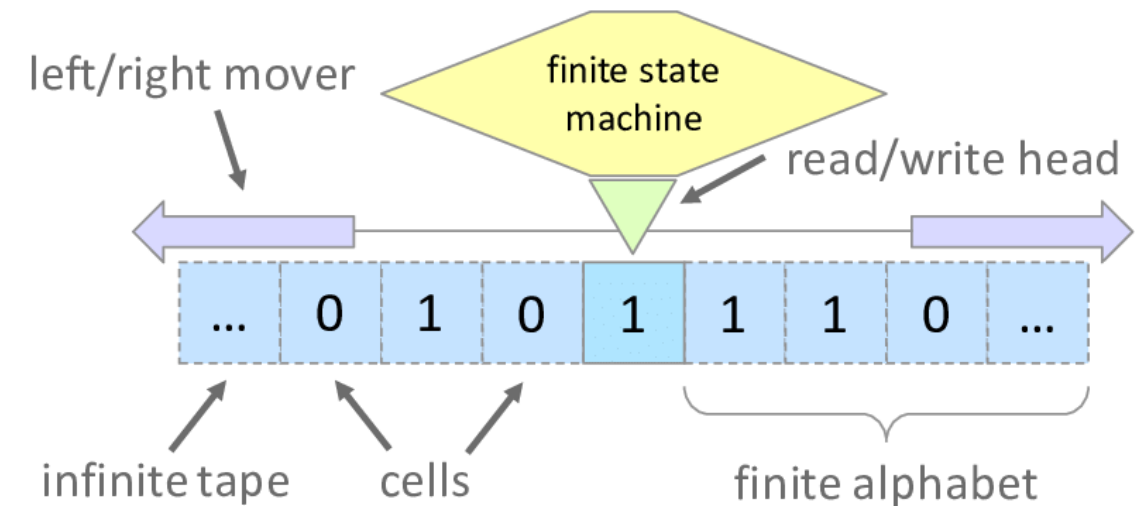
An infinite **tape**, made up of individual **cells**

Each cell contains a **symbol**, from a finite **alphabet**

A **read/write head**, which can move in each direction (one cell at time)

A **finite state machine**:

- The machine starts in an *initial state*
- Each *state transition* is governed by the input symbol and the current state
- The *next state* is stored into a register
- The *output* is written to the cell
- Then the head moves (i.e. *Left, None, Right*)



Turing Machine (A. Turing, 1937)

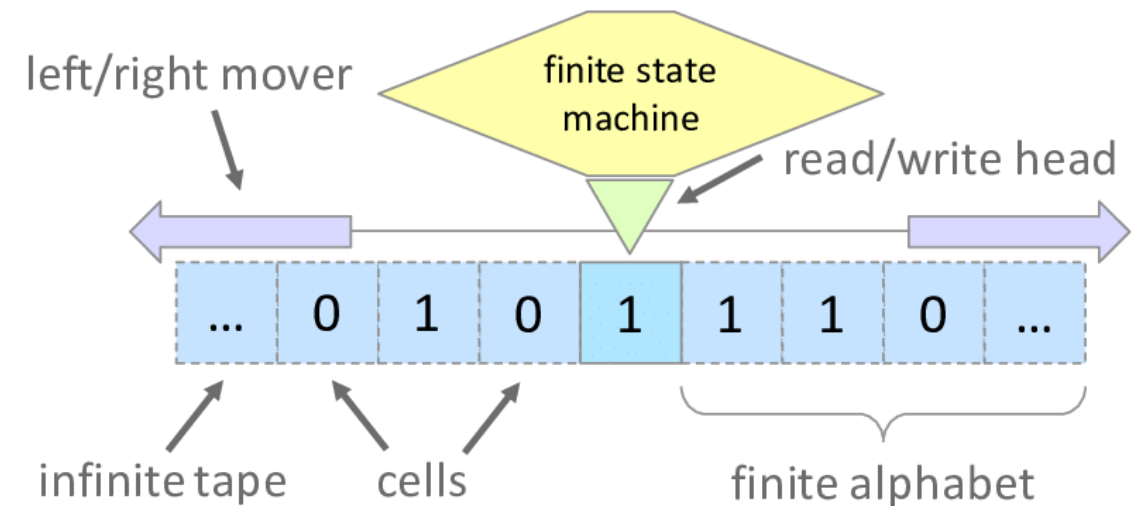
- What is the meaning of this?

The Turing Machine is a mathematical model of a physical computing device

Any given problem for which there is a Turing Machine that computes the solution is clearly computable by a physical machine

Is the vice-versa also true?

(Whenever a problem is computable by a physical machine, does it exist a Turing Machine for it?)



Church-Turing Thesis



- A possible formulation* (from Wikipedia):

*"Every '**function which would naturally be regarded as computable**' can be computed by a Turing machine."*

The vagueness in the above sentence gives raise to different interpretations. One of these (though not entirely equivalent) is (from Wikipedia):

*"Every '**function that could be physically computed**' can be computed by a Turing machine." ***

** Caution: there is no such a thesis in the original writings of either author.
Its formulation could be extrapolated from both, hence the attribution (made by others)*

*** Quantum computation shatters complexity theory, but is (almost) innocuous to computability theory*

Can machines think? (the Turing Test)

■ **The Imitation Game** (*textual interactions only*)

"A man (A), a woman (B),
and an interrogator (C) who may be of either sex.

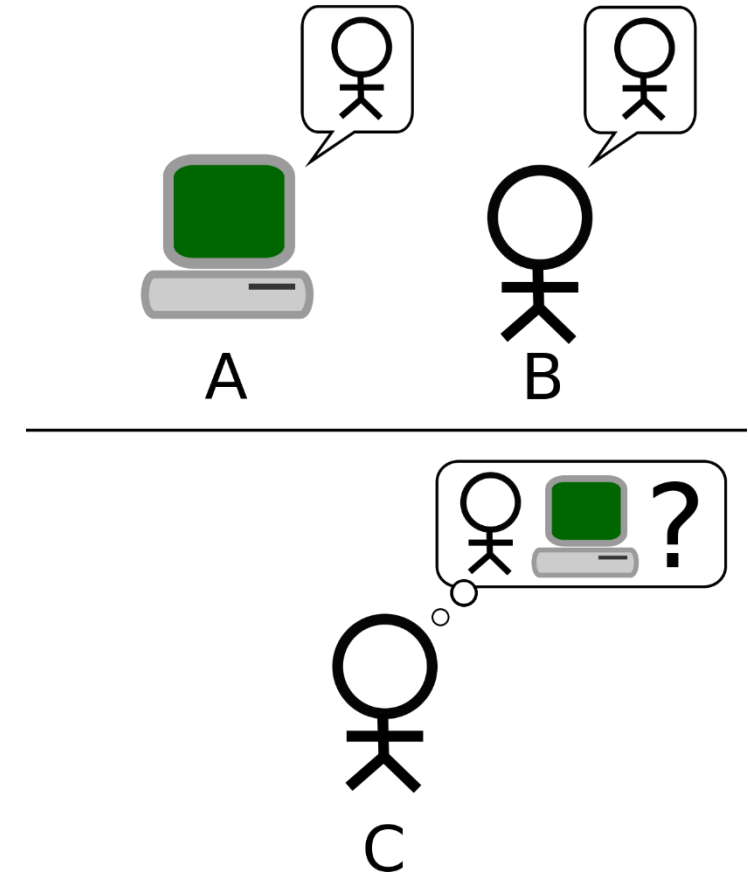
The interrogator stays in a room apart from the other two.

The object for the interrogator is to determine
which of the other two is the man and which is the woman.

The interrogator is allowed to put **questions** to A and B.
[...]

We now ask the question,
'What will happen when **a machine** takes the part of A in this game?'
[...]

These questions replace our original, 'Can machines think?' "



[Image from Wikipedia]

[Turing, A., Computing Machinery and Intelligence, 1950]

An aside question: Are we machines?

TURING TEST EXTRA CREDIT:
CONVINCE THE EXAMINER
THAT HE'S A COMPUTER.



YOU KNOW, YOU MAKE
SOME REALLY GOOD POINTS.
/ I'M ... NOT EVEN SURE
WHO I AM ANYMORE.

[Image from <https://xkcd.com/329/>]

A short summary of this course

Artificial Intelligence: *Part 1*

- *Reasoning with symbols*

Propositional logic, first-order logic, logic programming

Representation: language and semantics (*symbolic logic formulas and their meaning*)

Inference: entailment, a semantic relation among symbolic formulas

Automation: can machines compute entailment?

Plausible reasoning: beyond the scope of logical entailment

Artificial Intelligence: *Part 2*

- *Reasoning with numbers*

Machine learning (*in the probabilistic way*)

Representation: probability, random variables, graphical models

Inference: answers to queries from joint probability distributions

Causality: can graphical models be interpreted as *causal* models, as well?

Supervised learning: learning from *completely observed* and well-formed data items

Unsupervised learning: when some parts of data items are either *missing* or *hidden*

Reinforcement learning: learning from experience (even *online*)

Artificial Intelligence: *KIRO*

- ***Please register on KIRO***

<https://elearning.unipv.it/course/view.php?id=10171>

- Access all course materials (video recordings included)
- Messages & news about the course